On the connectedness and disconnectedness of the Julia set for the Hénon map

(Joint work with Yutaka Ishii)

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集中講義@京大人環

荒井迅

Zin Arai

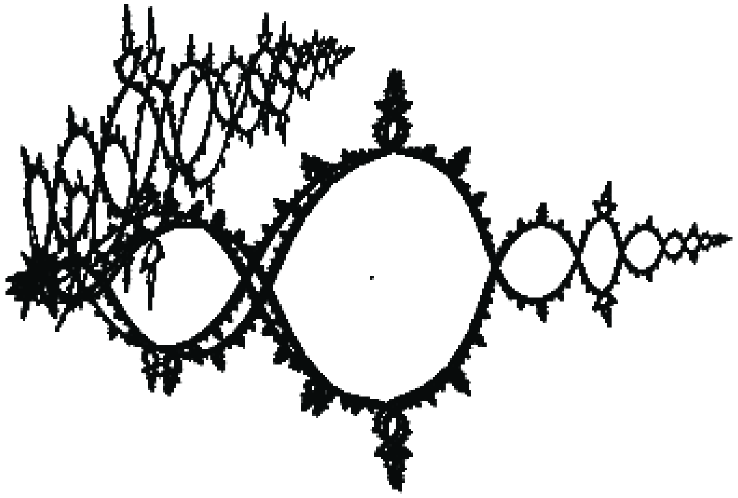
Institute of SCIENCE TOKYO

CREST

科研費 KAKENHI

Contents

Today's Goal: A rigorous algorithm for proving the disconnectedness of the Julia set for the complex Hénon map



Julia Set
for the Quadratic Map

The Complex Quadratic Map

The complex quadratic map

$$P_c : \mathbb{C} \to \mathbb{C} : z \mapsto z^2 + c \qquad (c \in \mathbb{C} \text{ is the parameter})$$

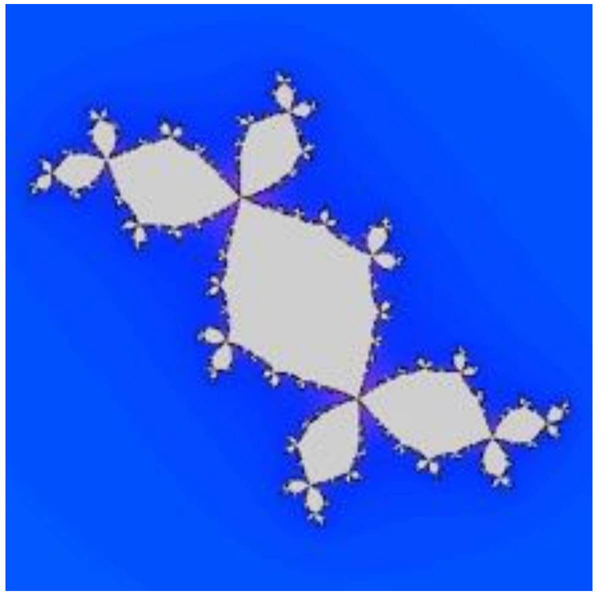
The simplest nonlinear dynamics

$$P_c^n := P_c \circ P_c \circ \cdots \circ P_c \text{ (n-times)}$$

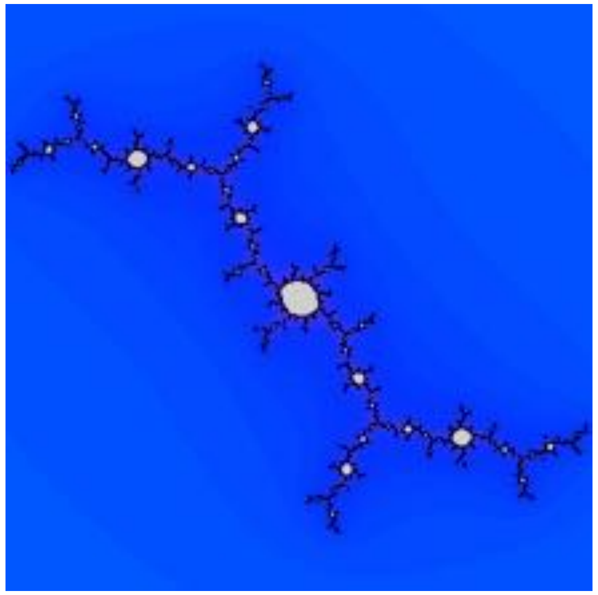
Gaston Julia

The filled Julia set and the Julia set

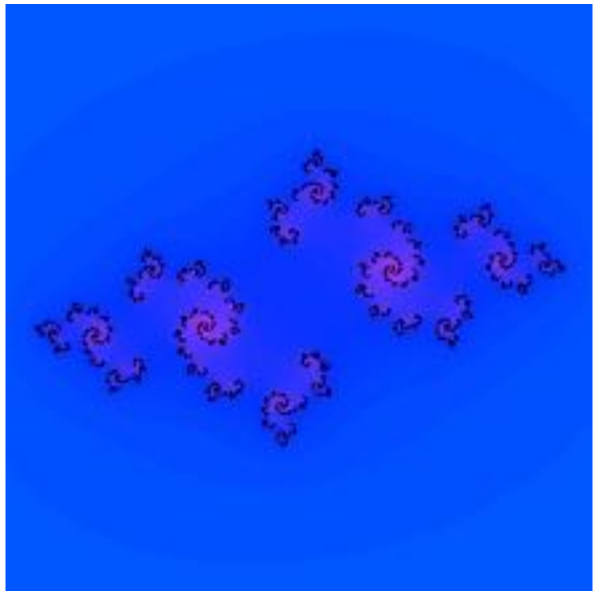
$$K_c := \{z \in \mathbb{C} \mid \{P_c^n(z)\} \text{ is bounded}\} \qquad J_c := \partial K_c$$



$c = -0.125475 + 0.7494i$



$c = -0.157414272428 + 1.033054947853i$



$c = -0.810000061989 - 0.344999969006i$

Julia Set
for the Hénon Map

The Hénon map

$$f_{a,c} : (x, y) \mapsto (x^2 + c - ay, x)$$

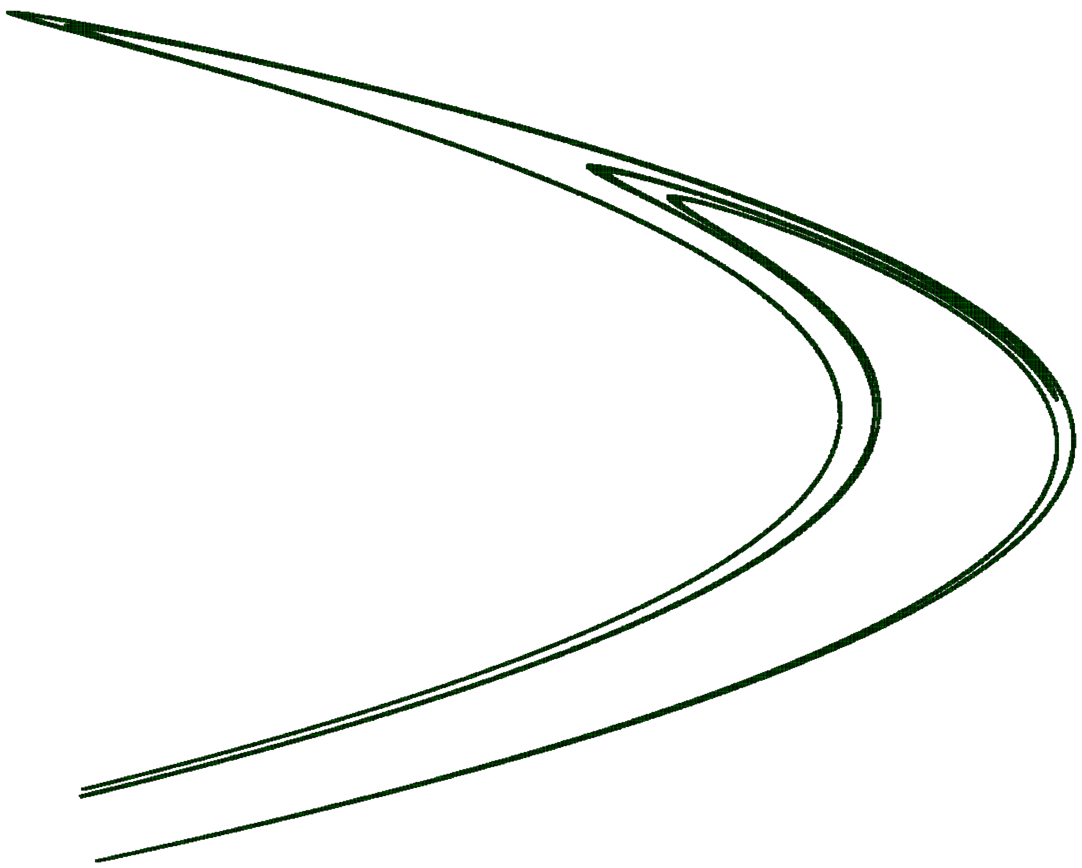
No critical point

- When $a = 0$, $f_{a,c}$ is the quadratic map.
- For $a \neq 0$, the map $f_{a,c}$ is a diffeomorphism and hence there is no critical point.

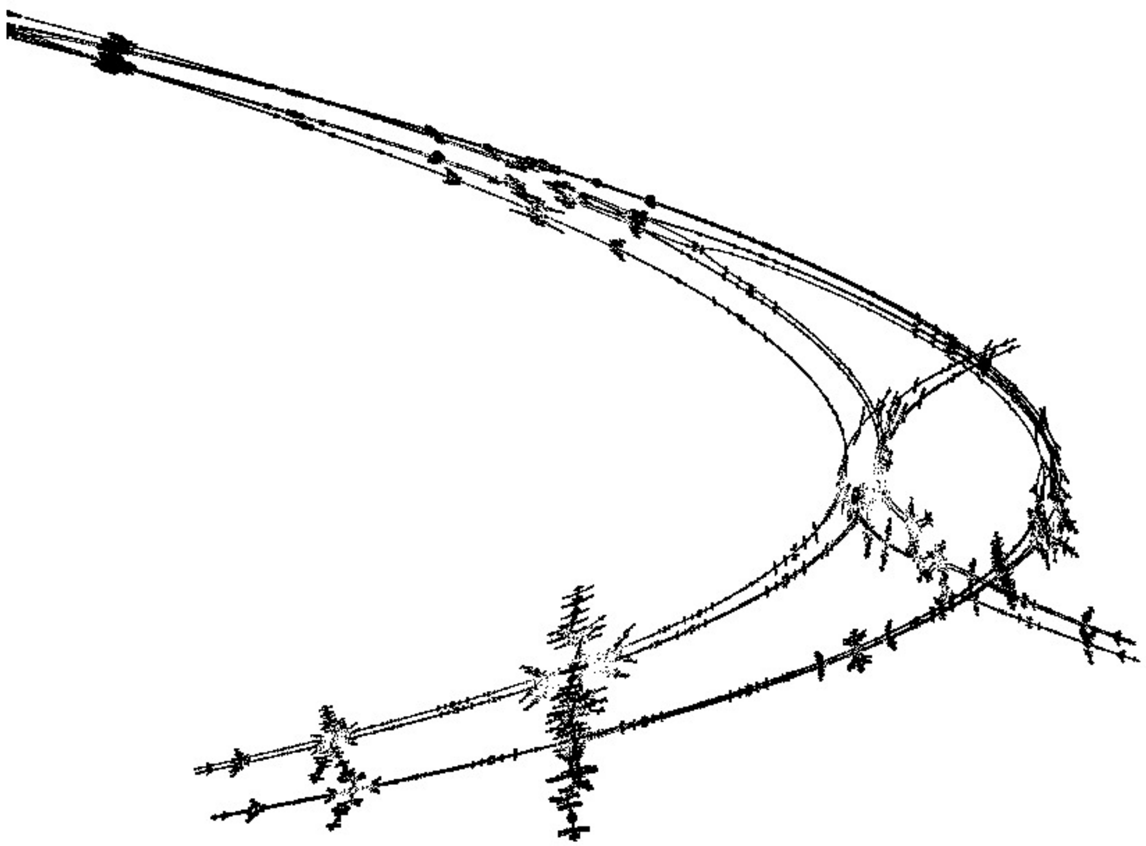
The Julia Set for the Hénon map

- $K^{\pm} := \{(x, y) \mid f^{\pm n}(x, y) \text{ remain bonded}\}$
- $K := K^{+} \cap K^{-}$: the filled Julia set
- $J^{\pm} := \partial K^{\pm}$
- $J := J^{+} \cap J^{-}$: the Julia set

Complex Picture

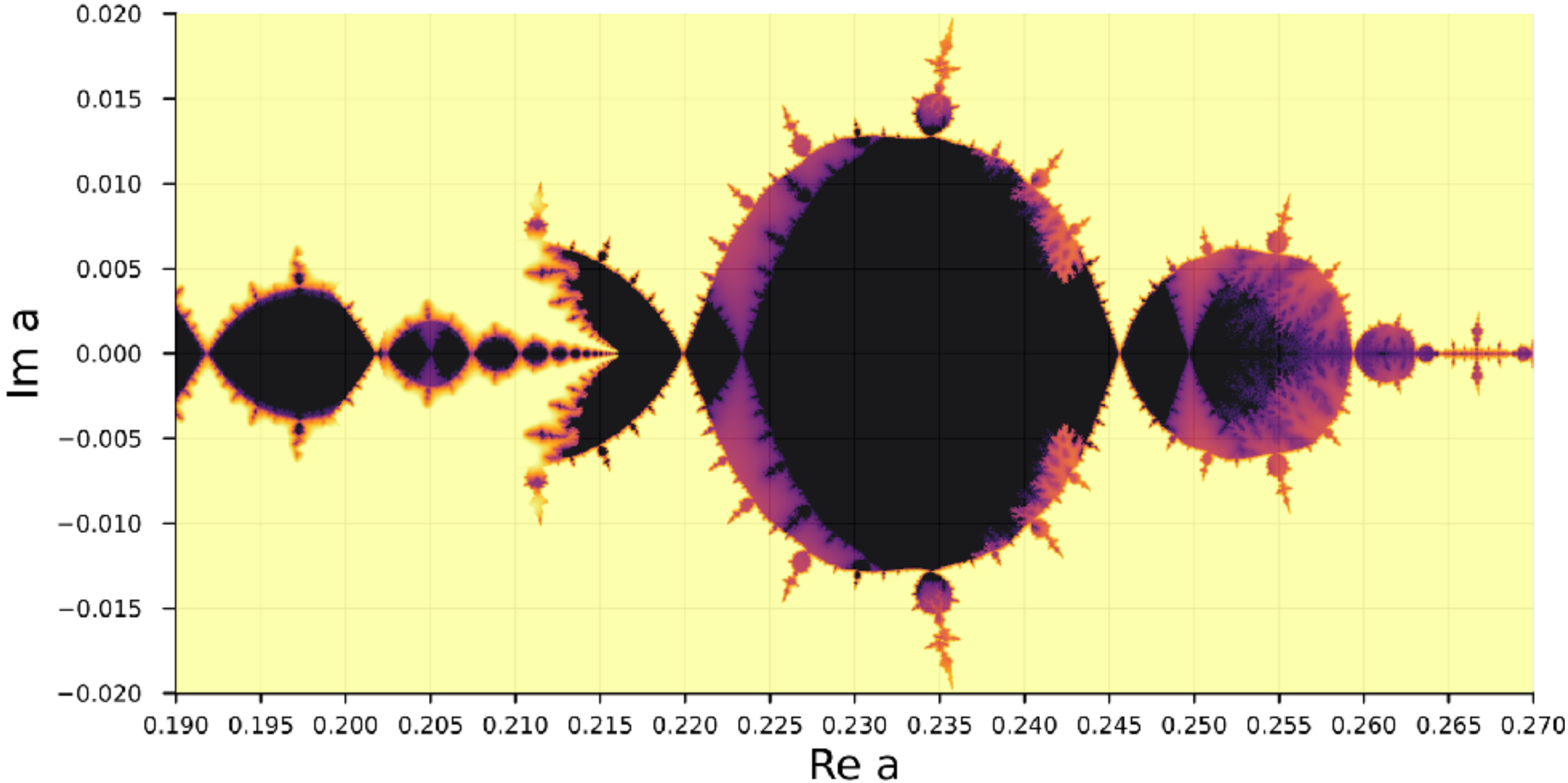


the strange attractor
(real Hénon map)



its complexification
(by S. Ushiki)

Complex Parameter Space

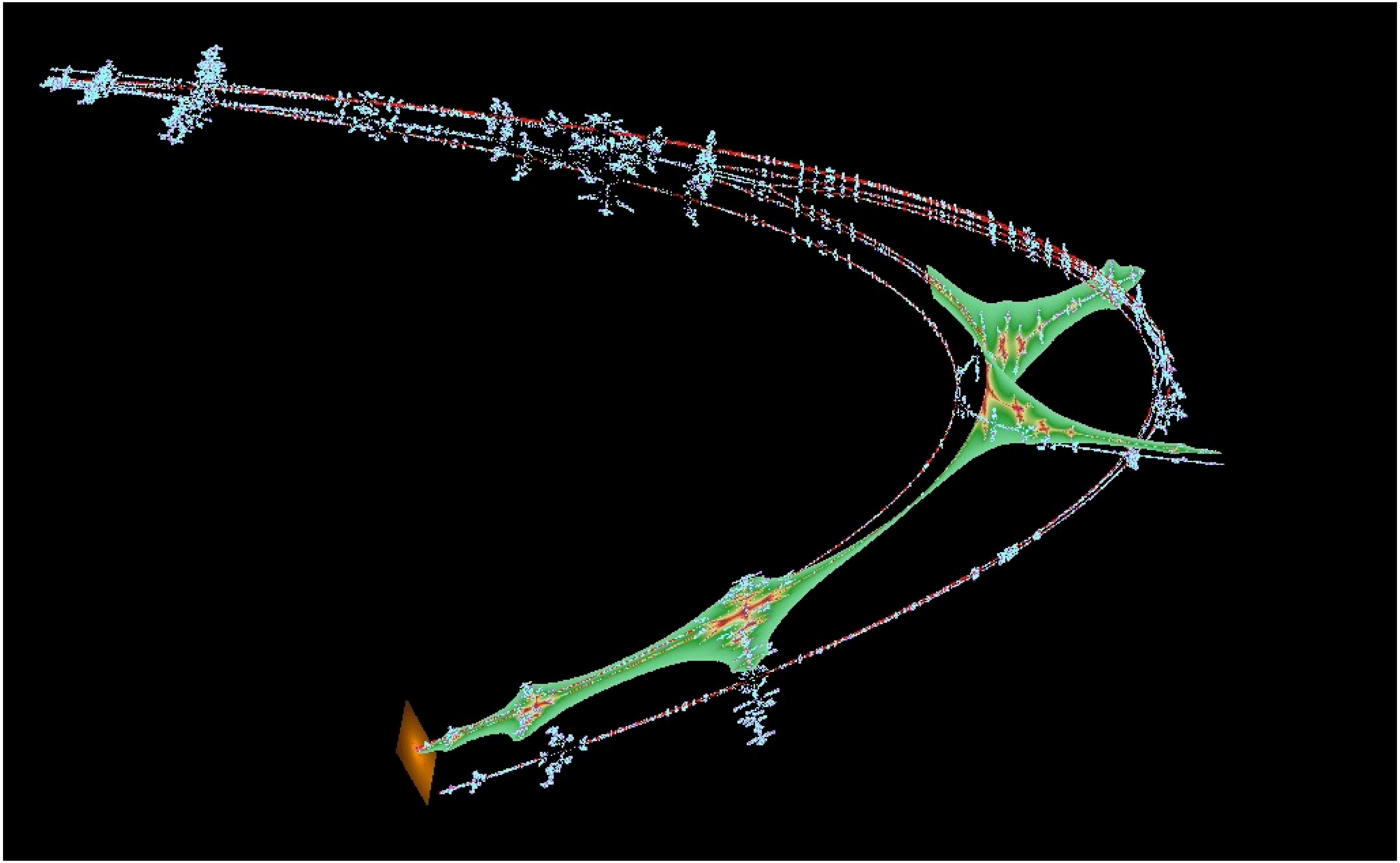


Non-rigorous picture of parameter space with c = -1.29

Black: connectedness locus

Yellow: full-shift locus

Complex Unstable Manifold

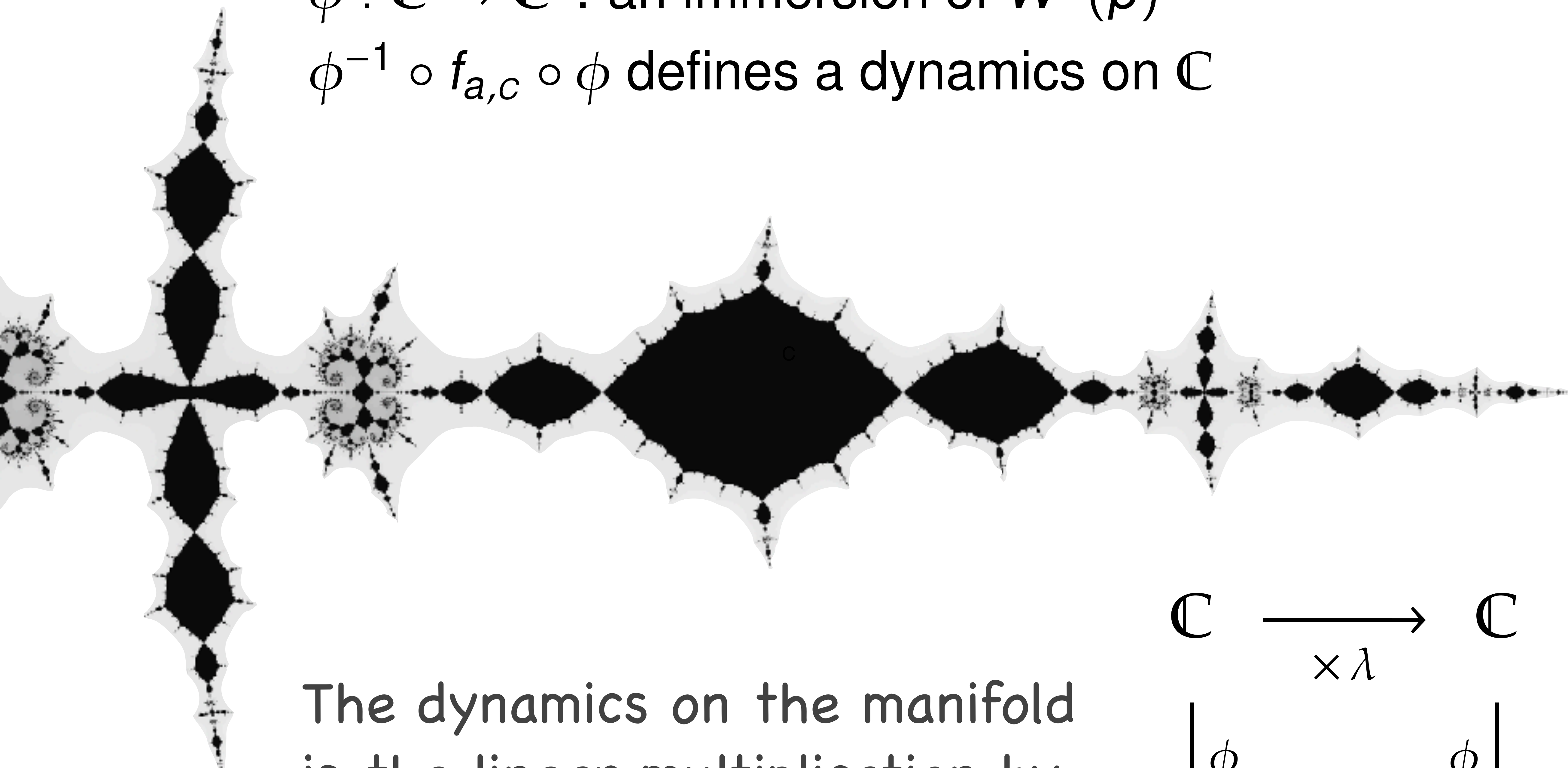


Complex (un)stable mfds by Shigehiro Ushiki

Julia Slice on the Unstable Manifold

$\phi : \mathbb{C} \to \mathbb{C}^2$: an immersion of $W^u(p)$

$\phi^{-1} \circ f_{a,c} \circ \phi$ defines a dynamics on $\mathbb{C}$



The dynamics on the manifold is the linear multiplication by the unstable eigenvalue λ

$$\begin{tikzcd} \mathbb{C} \arrow[r, "\times \lambda"'] \arrow[d, "\phi"] & \mathbb{C} \arrow[d, "\phi"'] \\ \mathbb{C}^2 \arrow[r, "f_{a,c}"] & \mathbb{C}^2 \end{tikzcd}$$

The Green function
and the maximum principle

Green Function

We define the Green function for $f = f_{a,c}$ by

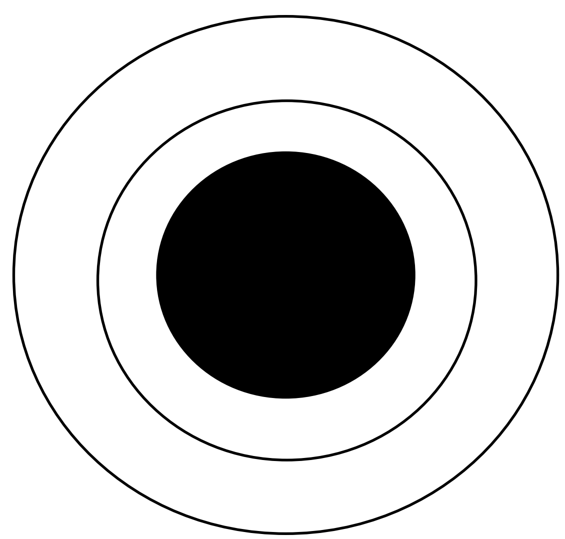
$$G^+(x, y) = \lim_{n\to\infty} \frac{1}{2^n} \log^+ \|f^n(x, y)\|$$

where $\log^+ z = \max\{\log z, 0\}$.

- G^+ is (pluri)harmonic on $\mathbb{C}^2 \setminus K^+$.
- $G^+ = 0$ on K^+.
- G^+ is (pluri)subharmonic on the entire $\mathbb{C}^2$.

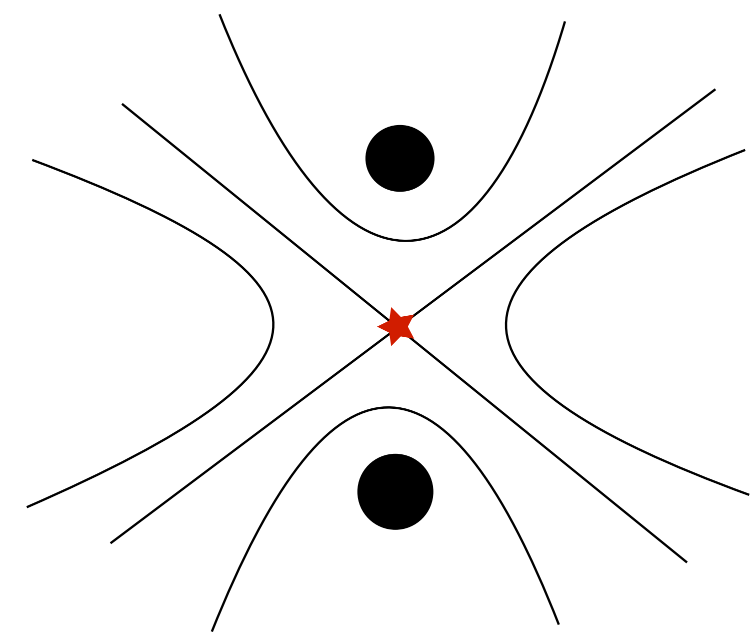
Critical Point of Green Function

connected



No critical point

disconnected



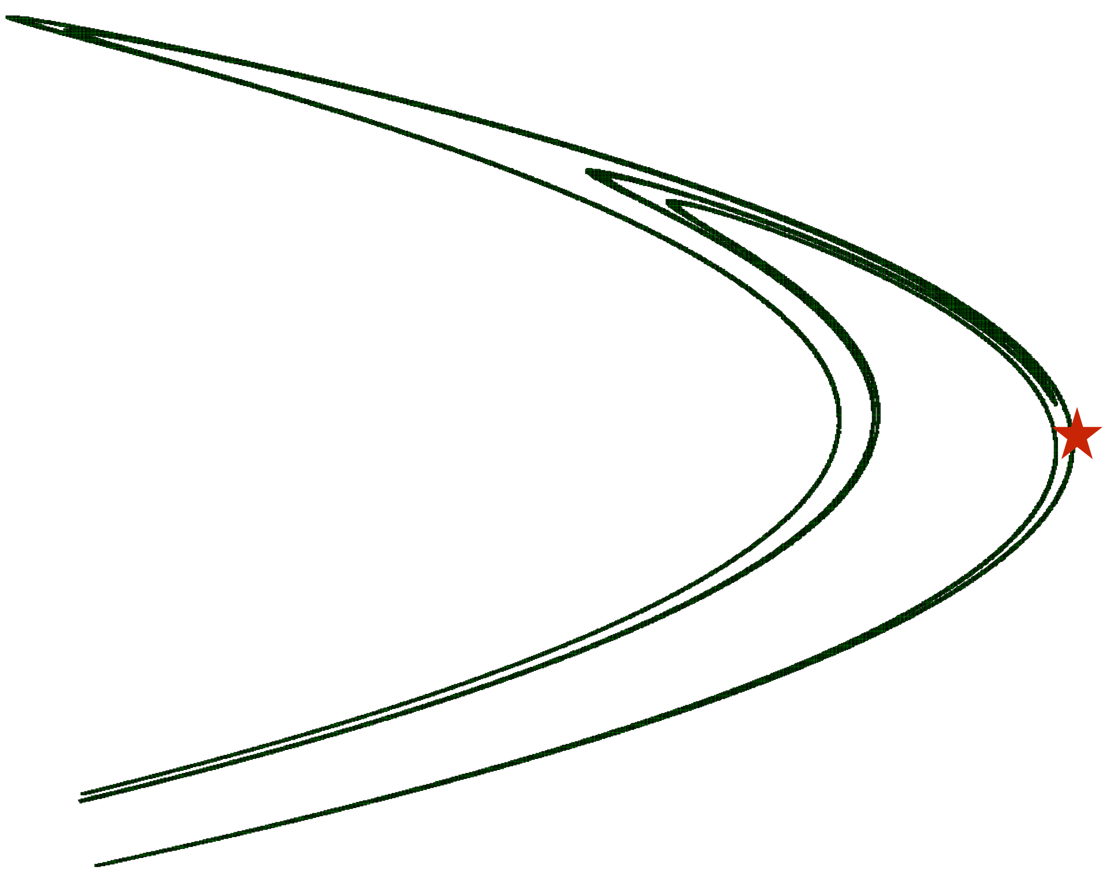
Critical point appears

Proposition (1-dim case)

Consider $P_c(z) = z^2 + c$. If z is a critical point of G^+ which is not contained in the filled Julia set, then $\exists m \geq 0$ such that $P_c^m(z) = 0$. Therefore,

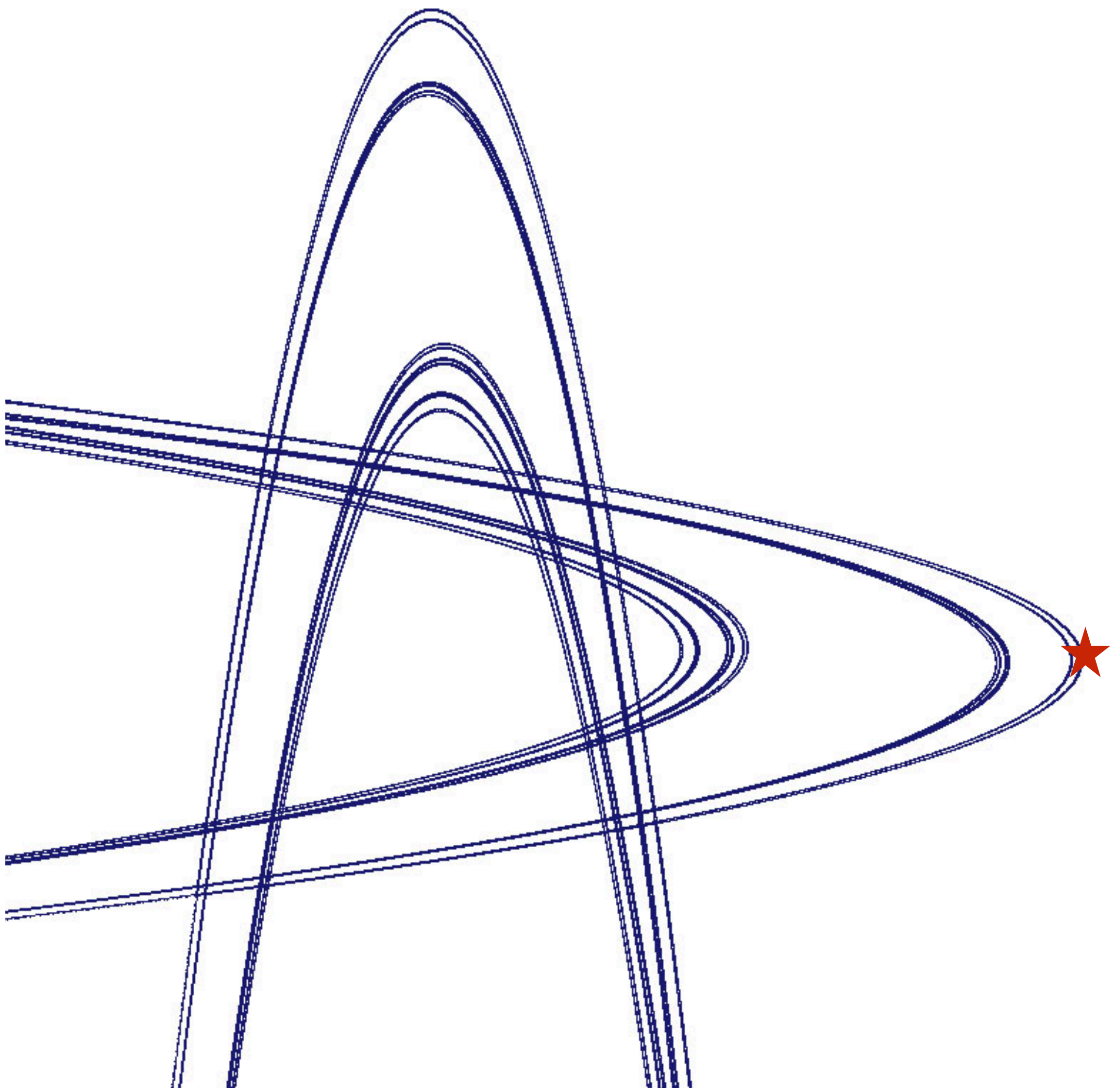
$$\exists \text{ a crit. point in } \mathbb{C} \setminus K_c \quad \Rightarrow \; 0 \notin K_c \; \Rightarrow \; K_c \text{ is disconnected.}$$

"Critical Points" and Turning Points



"Strange" attractor

* The orbit of ★ is bounded
* Julia set looks connected on $\mathbb{R}^2$ (but it's disconnected in $\mathbb{C}^2$)



Hyprebolic horseshoe

* The orbit of ★ is unbounded
* Julia set is totally disconnected

A Criterion for 2D

Theorem (Bedford-Smillie, Dujardin)

Let p be a saddle point of f. Then the followings are equivalent.

- *J_f is connected*
- *the restriction of G^+ on $W^u(p) \setminus K_f$ has no critical point.*
- *$W^u(p) \cap K_f$ has no compact component*

Let $\phi : \mathbb{C} \to \mathbb{C}^2$ the analytic function for $W^u(p)$, that is,

$$\phi(0) = p, \quad \phi'(0) = \lambda, \quad f(\phi(z)) = \phi(\lambda(z))$$

where λ is the unstable eigen value at p.

To prove the disconnectedness of J_f, it suffices to show the existence of a critical point of $G^+ \circ \phi$.

A numerical search for the critical point is relatively easy, but...

Bound for the unstable manifold

Let $\phi(z) = (x(z), y(z))$ and

$$\begin{aligned} x(z) &= s_0 + s_1 z + s_2 z^2 + s_3 z^3 + \cdots \\ y(z) &= t_0 \; + t_1 z \; + t_2 z^2 \; + t_3 z^3 + \cdots \end{aligned}$$

be the Taylor expansion of x and y.

Then (s_0, t_0) and (s_1, t_1) is determined by p and the eigenvector.

For $n > 1$, it follows from $f(\phi(z)) = \phi(\lambda(z))$ that

$$\begin{aligned} s_n &= \frac{\sum_{i=1}^{n-1} s_i s_{n-i}}{\lambda^n + \frac{a}{\lambda^n} - 2s_0} \\ t_n &= s_n / \lambda^n. \end{aligned}$$

We also have a rigorous estimate of the reminder term.

Bounds for the Green function

Let

$$R_0 = \frac{1 + |a| + \sqrt{(1 + |a|)^2 + 4|c|}}{2}$$

$$R_1 = \max\{R_0, |a| + \sqrt{|a|^2 + 2|c|}\}$$

and define

$$V_{R_0} = \left\{(x, y) \in \mathbb{C}^2 : |x| \geq R_0,\ |x| \geq |y|\right\}$$

$$V_{R_1} = \left\{(x, y) \in \mathbb{C}^2 : |x| \geq R_1,\ |x| \geq |y|\right\}.$$

Lemma

1. $f^n(x_0, y_0) \in V^+_{R_0} \Rightarrow G^+(x_0, y_0) \leq \frac{1}{2^n}(\log|x_n| + \log 2)$,
2. $f^n(x_0, y_0) \in V^+_{R_1} \Rightarrow G^+(x_0, y_0) \geq \frac{1}{2^n}(\log|x_n| - \log 2)$.

Maximum (Minimum) Principle

Lemma

Let $C > 0$ and γ be a loop in $\mathbb{C}$ such that

$$G^+(\phi(z)) > C$$

for all $z \in \gamma$ and assume there is z_0 inside γ such that

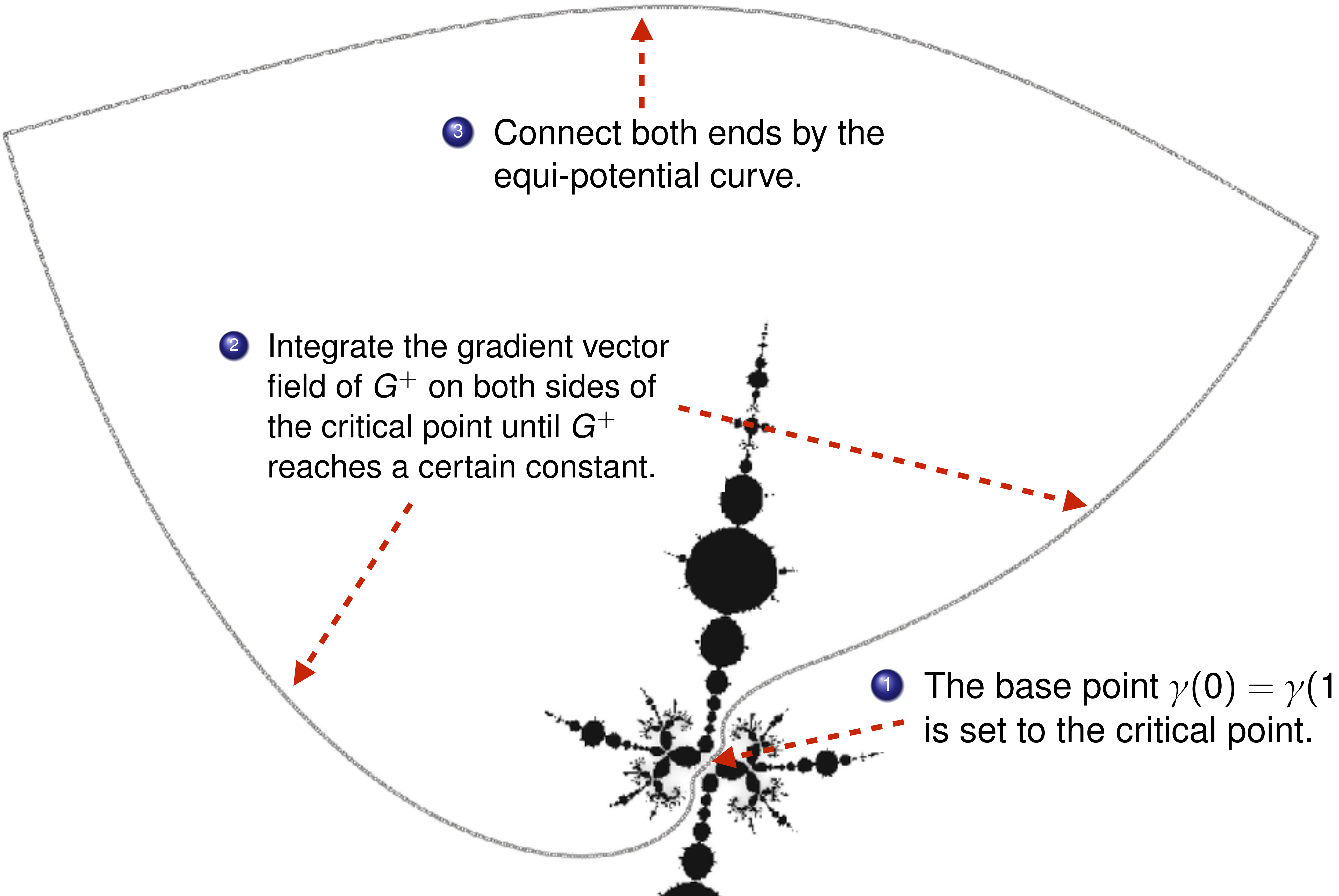
$$G^+(\phi(z_0)) < C.$$

Then $\phi(\gamma)$ should enclose a point of K_f.

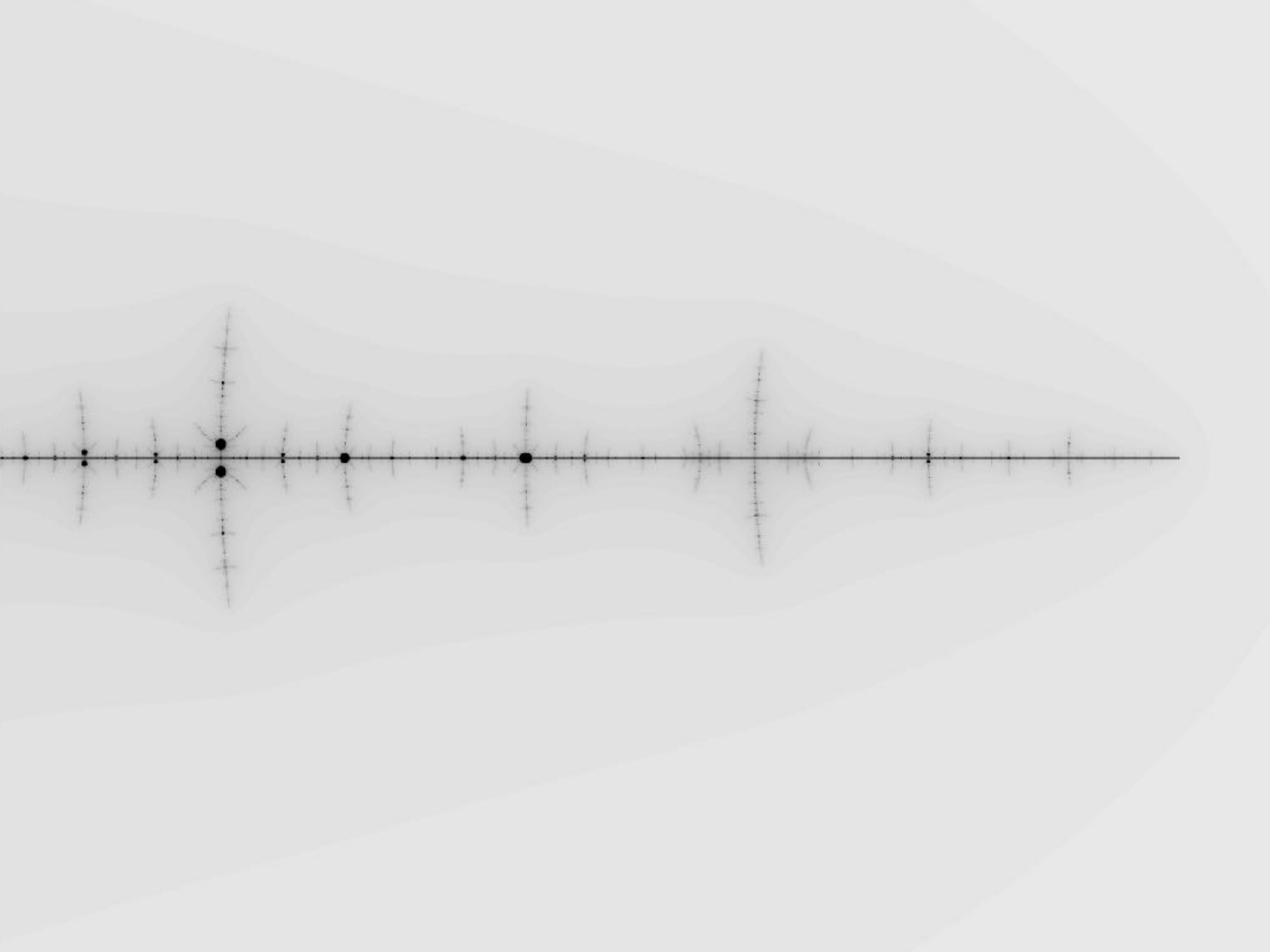
Theorem

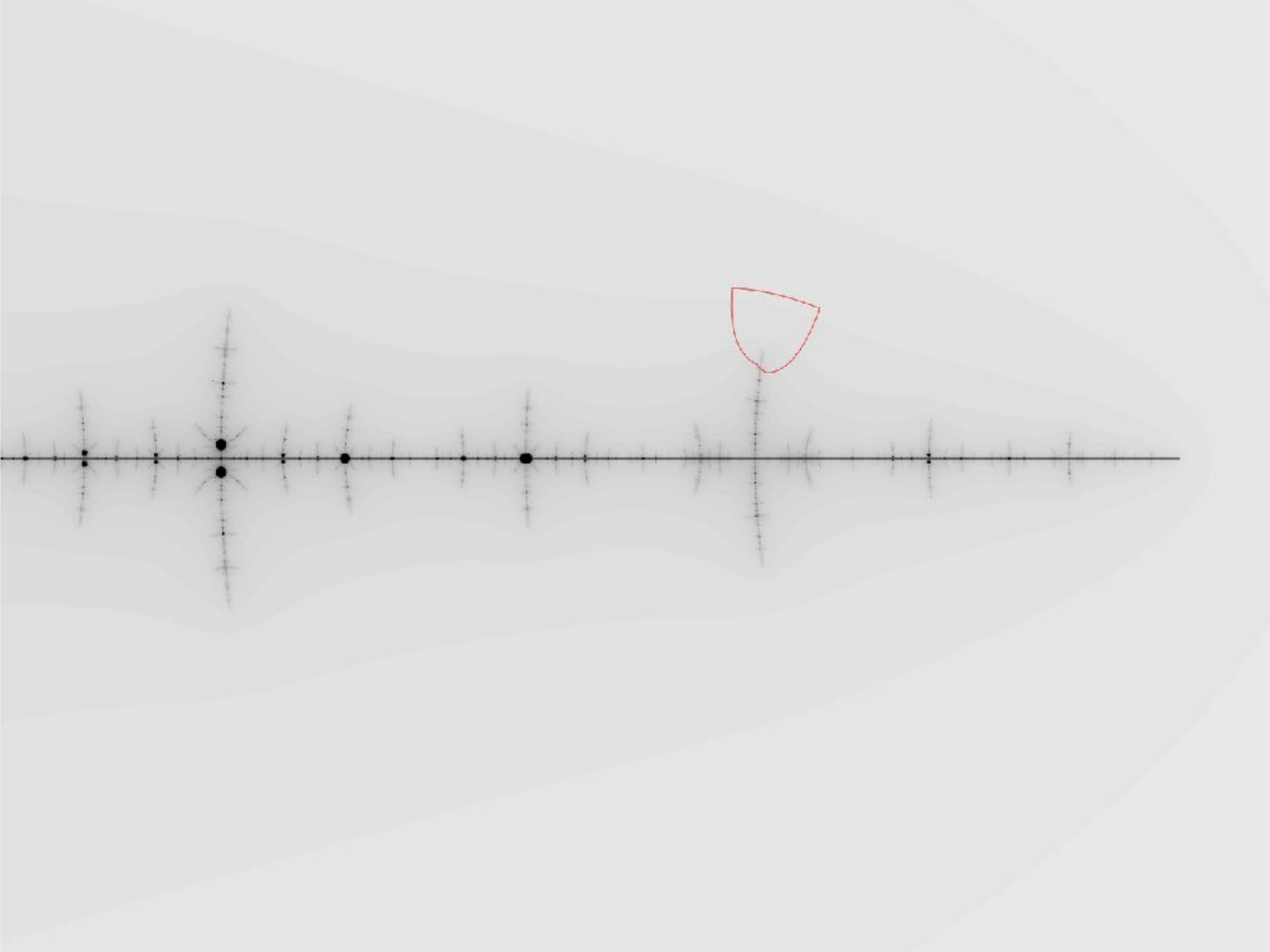
If there is a loop that satisfies the assumption of the Lemma and does not enclose the origin of $\mathbb{C}$, then J_f is disconnected.

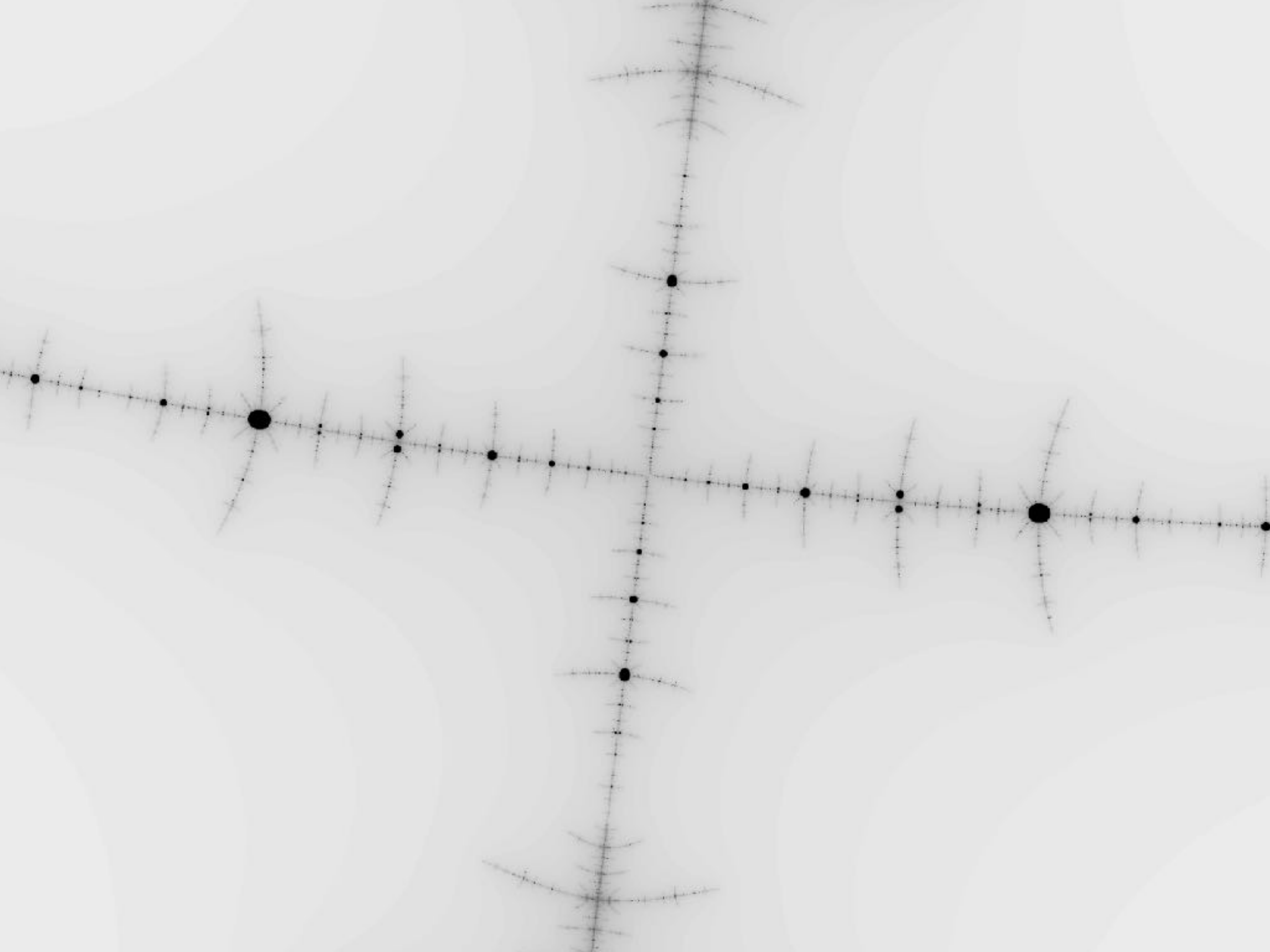
Steps to construct the loop

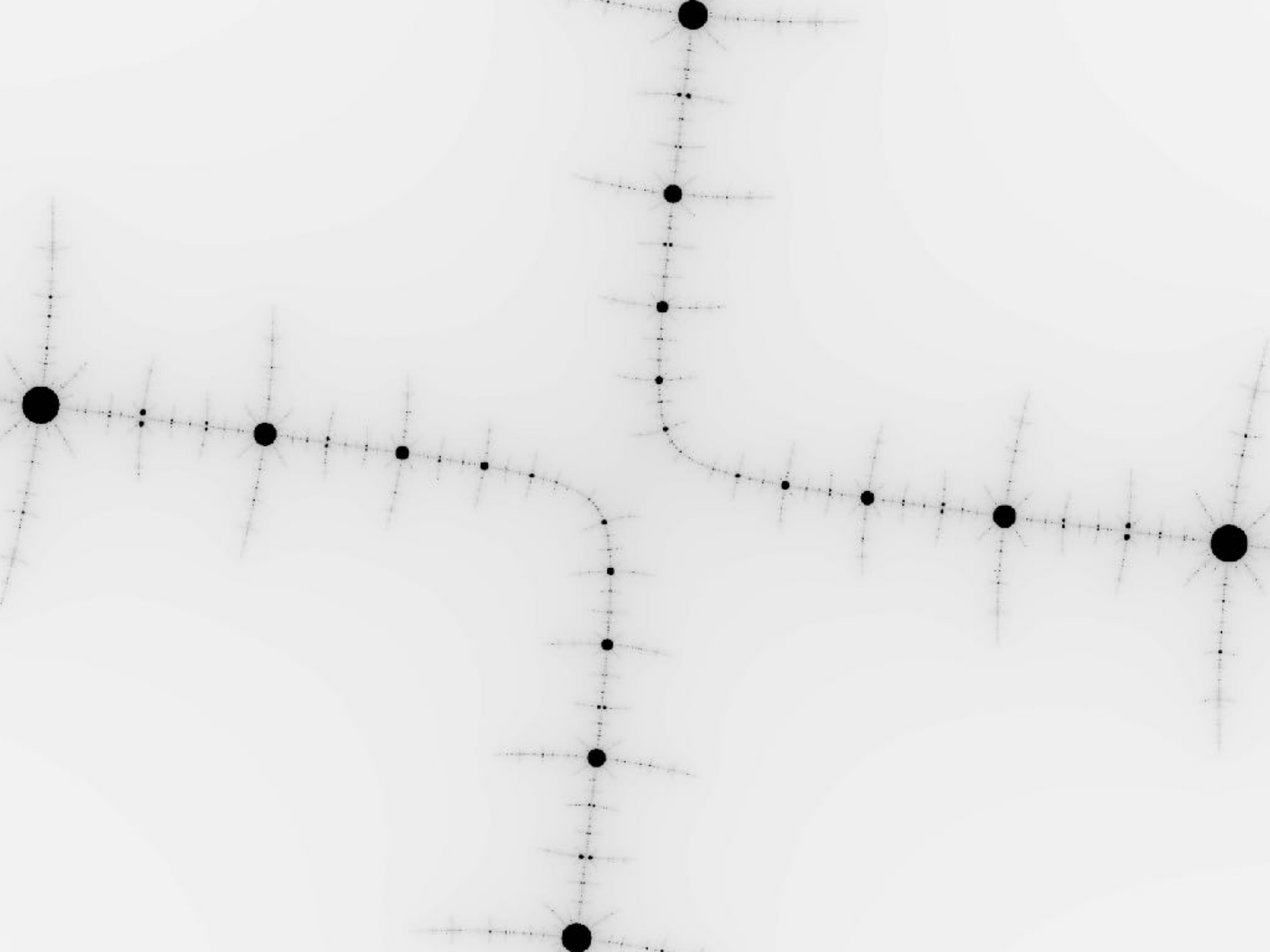


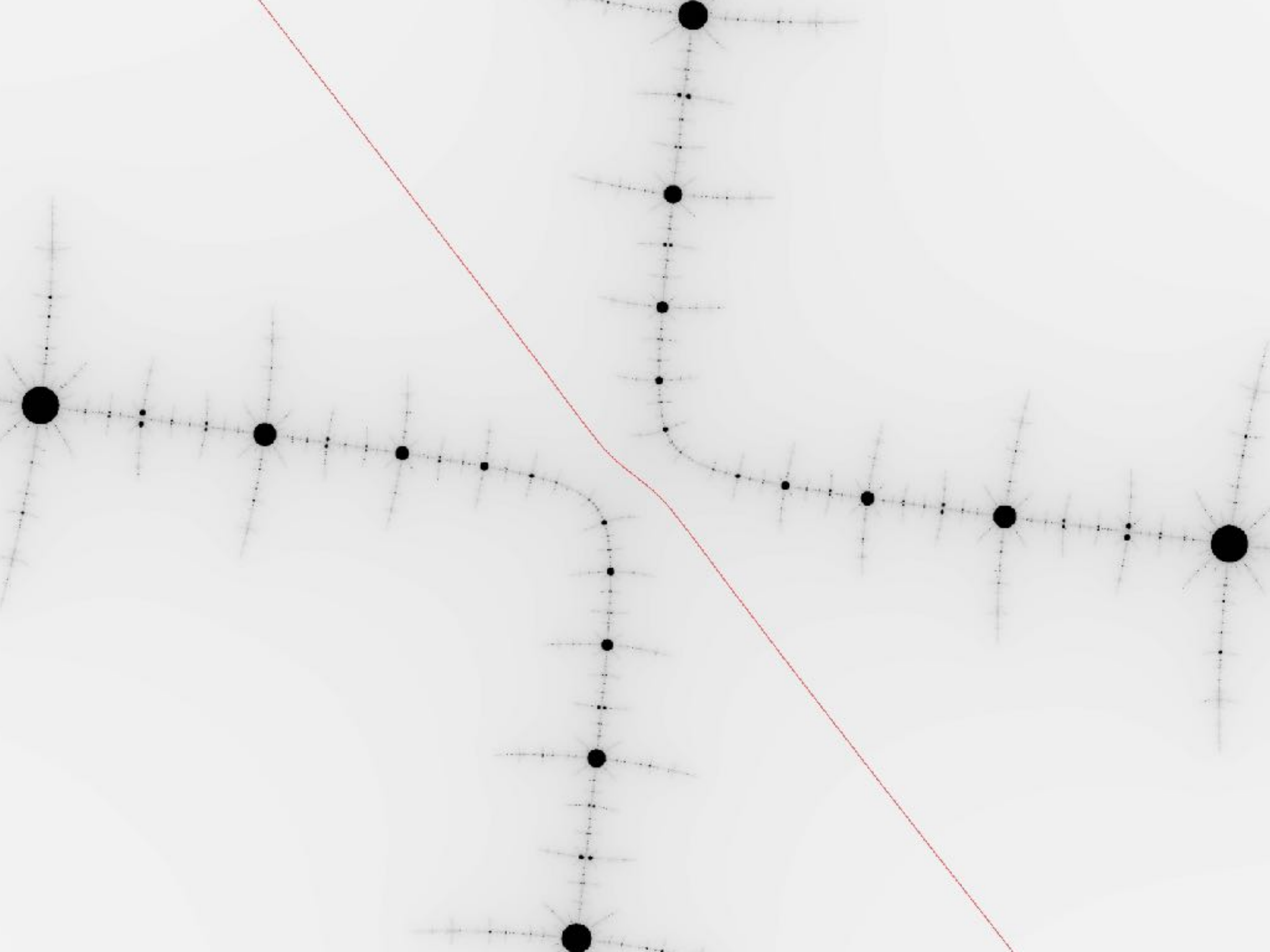
Examples

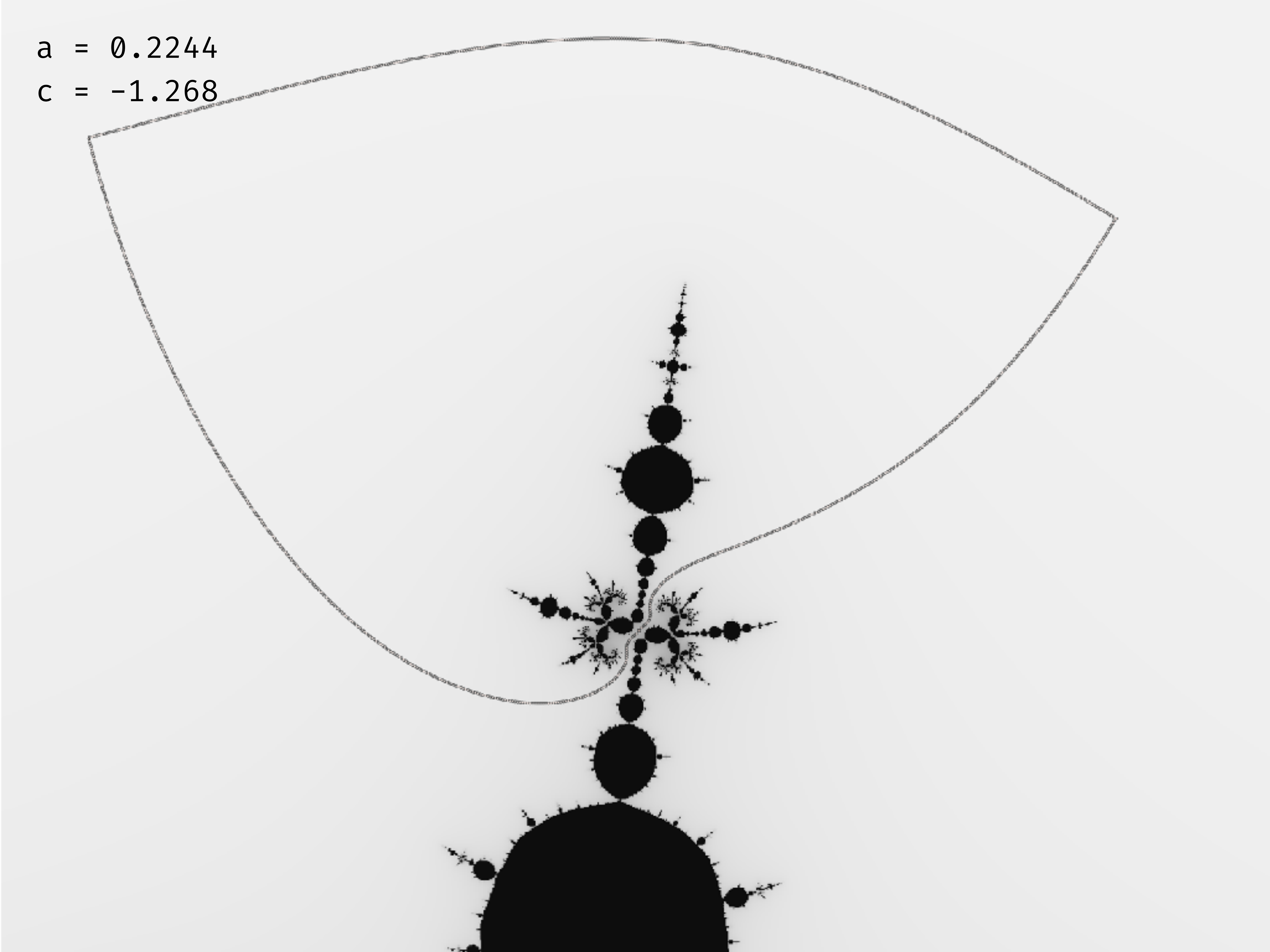






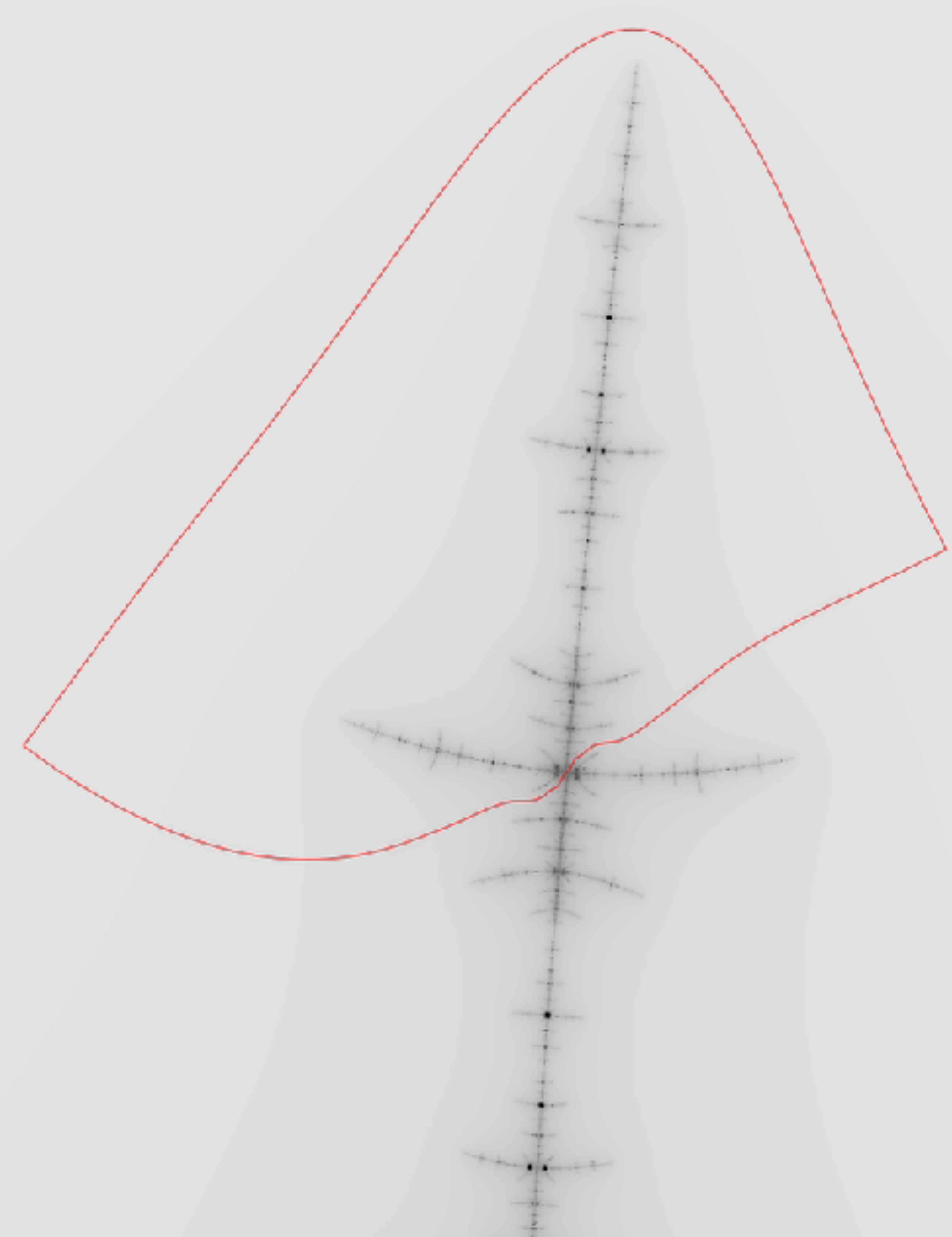


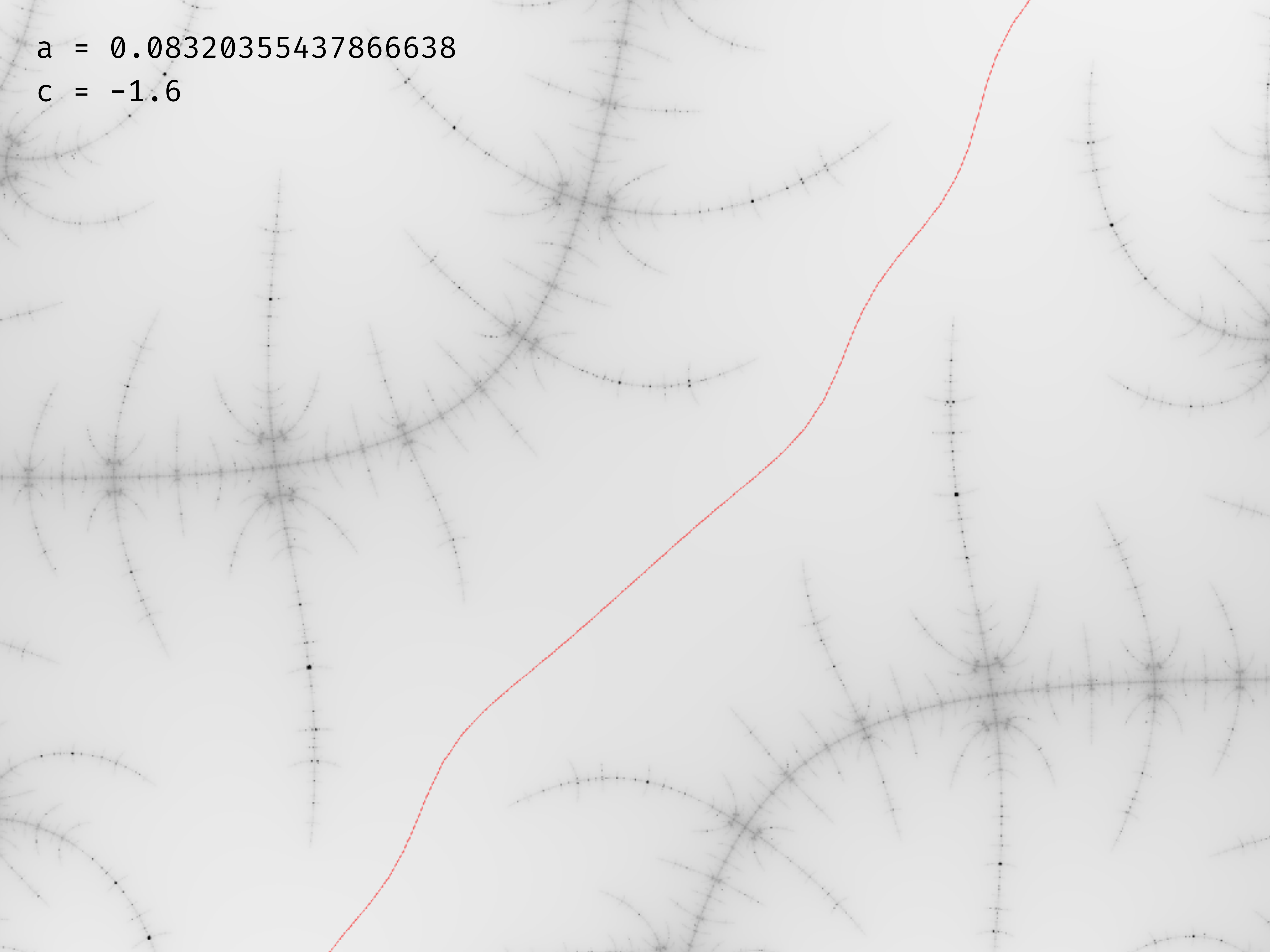


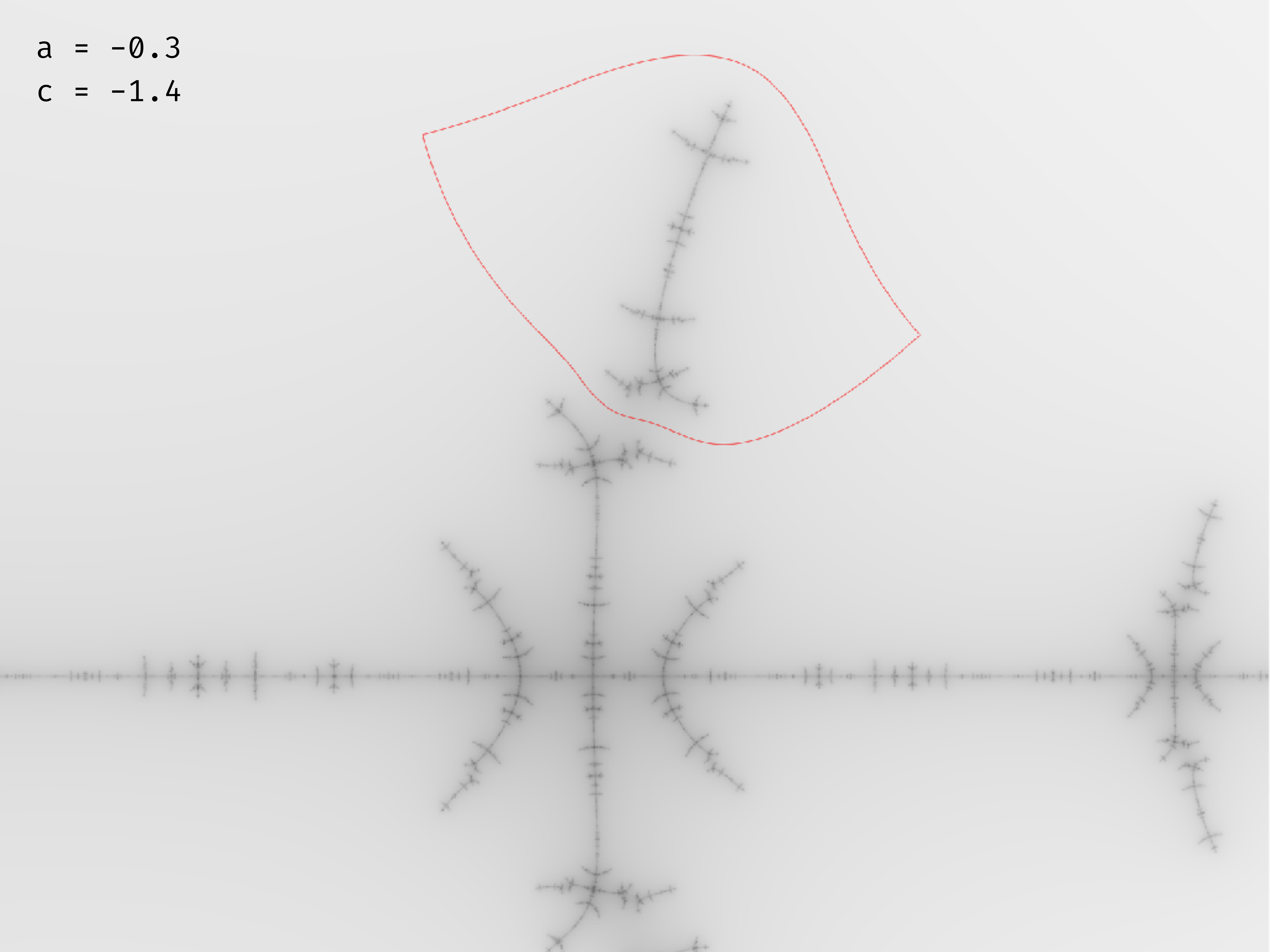

a = 0.2244
c = -1.268

a = 0.08320355437866638

c = -1.6




a = 0.08320355437866638
c = -1.6


a = -0.3
c = -1.4

Theorems and a conjecture

Theorem (connectedness): The Hénon map with (a, c) = (0.23, -1.93) is hyperbolic and non-planner with a connected Julia set.

Let $\mathcal{M} = \{(a,c) \in \mathbb{C}^2 \mid f_{a,c} \text{ has a connected Julia set}\}$

Theorem: $\mathcal{M} \cap \mathbb{R}^2$ is not connected.

Conjecture: $\mathcal{M}$ is not connected.